



**Progressive Education Society's Modern College of Engineering, Pune-05. (An Autonomous Institute
Affiliated to Savitribai Phule Pune University) Department of S.Y. B. Tech [AIDS]**

Assignment No: 10

SUBJECT: Java Programming

Name: Om Mane

Class: SY BTech (AIDS)

Section: A

Roll Number: 26040

Title / Problem Statement :

Develop a JavaFX application that provides a graphical user interface to perform basic integer arithmetic operations. the interface should include: a. two input fields labeled 'number 1' and 'number 2'. b. four buttons labeled '+', '-', '*', and '/'. c. a display field labeled 'result'.

```
Input :import javafx.application.Application;

import javafx.geometry.Insets;

import javafx.scene.Scene;

import javafx.scene.control.*;

import javafx.scene.layout.GridPane;

import javafx.stage.Stage;


public class Main extends Application {

    public void start(Stage stage) {

        TextField num1 = new TextField();

        TextField num2 = new TextField();

        TextField result = new TextField();

        result.setEditable(false);

        Button add = new Button("+");

        Button sub = new Button("-");

        Button mul = new Button("*");

        Button div = new Button("/");

        // Event Handling

        add.setOnAction(e -> {

            int a = Integer.parseInt(num1.getText());

            int b = Integer.parseInt(num2.getText());

            result.setText(String.valueOf(a + b));

        });
```

```
sub.setOnAction(e -> {  
  
    int a = Integer.parseInt(num1.getText());  
  
    int b = Integer.parseInt(num2.getText());  
  
    result.setText(String.valueOf(a - b));  
  
});
```

```
mul.setOnAction(e -> {  
  
    int a = Integer.parseInt(num1.getText());  
  
    int b = Integer.parseInt(num2.getText());  
  
    result.setText(String.valueOf(a * b));  
  
});
```

```
div.setOnAction(e -> {  
  
    try {  
  
        int a = Integer.parseInt(num1.getText());  
  
        int b = Integer.parseInt(num2.getText());  
  
        result.setText(String.valueOf(a / b));  
  
    } catch (ArithmeticException ex) {  
  
        result.setText("Cannot divide by zero");  
  
    }  
  
});
```

```
GridPane grid = new GridPane();  
  
grid.setPadding(new Insets(10));  
  
grid.setVgap(10);  
  
grid.setHgap(10);  
  
  
grid.add(new Label("Number 1"), 0, 0);
```

```
grid.add(num1, 1, 0);
```

```
grid.add(new Label("Number 2"), 0, 1);
```

```
grid.add(num2, 1, 1);
```

```
grid.add(add, 0, 2);
```

```
grid.add(sub, 1, 2);
```

```
grid.add(mul, 0, 3);
```

```
grid.add(div, 1, 3);
```

```
grid.add(new Label("Result"), 0, 4);
```

```
grid.add(result, 1, 4);
```

```
Scene scene = new Scene(grid, 300, 250);
```

```
stage.setTitle("Simple Calculator");
```

```
stage.setScene(scene);
```

```
stage.show();
```

```
}
```

```
public static void main(String[] args) {
```

```
    launch(args);
```

```
}
```

```
}
```

Output :

File Edit Selection View Go Run Terminal Help

assignment_10

EXPLORER

ASSIGNMENT_10

Main.class

Main.java 9+

Main.java > Language Support for Java(TM) by Red Hat > Main > start(stage)

```
1 import javafx.application.Application;
2 import javafx.geometry.Insets;
3 import javafx.scene.Scene;
4 import javafx.scene.control.*;
5 import javafx.scene.layout.GridPane;
6 import javafx.stage.Stage;
7
8 public class Main extends Application {
9
10     public void start(Stage stage) {
11
12         TextField num1 = new TextField();
13         TextField num2 = new TextField();
14         TextField result = new TextField();
15         result.setEditable(false);
16
17         Button add = new Button("+");
18         Button sub = new Button("-");
19         Button mul = new Button("*");
20         Button div = new Button("/");
21
22         // Event Handling
23         add.setOnAction(e -> {
24             int a = Integer.parseInt(num1.getText());
25             int b = Integer.parseInt(num2.getText());
```

Simple Calculator

Number 1 10

Number 2 5

+ -

* /

Result 15

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS D:\College\College Coding\Java\assignment_10> javac --module-path "C:\javafx-sdk-26\lib" --add-modules javafx.controls Main.java

PS D:\College\College Coding\Java\assignment_10> java --module-path "C:\javafx-sdk-26\lib" --add-modules javafx.controls Main

WARNING: A restricted method in java.lang.System has been called

WARNING: java.lang.System::load has been called by com.sun.glass.utils.NativeLibLoader in module javafx.graphics (file:/C:/javafx-sdk-26/lib/javafx.graphics.jar)

WARNING: Use --enable-native-access=java:fx to avoid a warning for callers in this module

WARNING: Restricted methods will be blocked in a future release unless native access is enabled

Ln 28, Col 1 Spaces: 4 UTF-8 LF Java

37°C Sunny

11:36 AM 4/11/2026

File Edit Selection View Go Run Terminal Help assignment_10

EXPLORER

- ASSIGNMENT_10
 - Main.class
 - Main.java 9+

Main.java > Language Support for Java(TM) by Red Hat > Main > start(stage)

```
1 import javafx.application.Application;
2 import javafx.geometry.Insets;
3 import javafx.scene.Scene;
4 import javafx.scene.control.*;
5 import javafx.scene.layout.GridPane;
6 import javafx.stage.Stage;
7
8 public class Main extends Application {
9
10     public void start(Stage stage) {
11
12         TextField num1 = new TextField();
13         TextField num2 = new TextField();
14         TextField result = new TextField();
15         result.setEditable(false);
16
17         Button add = new Button("+");
18         Button sub = new Button("-");
19         Button mul = new Button("*");
20         Button div = new Button("/");
21
22         // Event Handling
23         add.setOnAction(e -> {
24             int a = Integer.parseInt(num1.getText());
25             int b = Integer.parseInt(num2.getText());
```

Simple Calculator

Number 1: 10
Number 2: 5
Result: 50

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS D:\College\College Coding\Java\assignment_10> javac --module-path "C:\javafx-sdk-26\lib" --add-modules javafx.controls Main.java
PS D:\College\College Coding\Java\assignment_10> java --module-path "C:\javafx-sdk-26\lib" --add-modules javafx.controls Main
WARNING: A restricted method in java.lang.System has been called
WARNING: java.lang.System::load has been called by com.sun.glass.utils.NativeLibLoader in module javafx.graphics (file:/C:/javafx-sdk-26/lib/javafx.graphics.jar)
WARNING: Use --enable-native-access=javaafx.graphics to avoid a warning for callers in this module
WARNING: Restricted methods will be blocked in a future release unless native access is enabled
```

OUTLINE
TIMELINE
PROJECTS
RUN CONFIGURATION
JAVA PROJECTS

Ln 28, Col 1 Spaces: 4 UTF-8 LF Java

17°C Sunny

11:36 AM 4/11/2026